

PeakSelf Passwordless Authentication Implementation Plan

1. Project Goal

Build a production - grade authentication system for PeakSelf using Magic Link login, Google OAuth, JWT access tokens, refresh tokens, Redis, and email delivery with Resend.

2. Technology Stack

Frontend: Next.js

Backend: NestJS

Database: PostgreSQL with Prisma

Cache: Redis

Queue: BullMQ

Email: Resend

Package Manager: pnpm

Monorepo: Turborepo

3. Phase 1 - Project Setup

Create a Turborepo with apps/web (Next.js) and apps/api (NestJS). Add shared packages for database, auth, and UI.

4. Phase 2 - Database Schema

Create Prisma models: User, Identity, MagicLinkToken, RefreshToken, Session, Device, and LoginHistory.

5. Phase 3 - Email Infrastructure

Configure Resend and create responsive email templates for login links.

6. Phase 4 - Magic Link Authentication

Generate single - use hashed tokens that expire in 10 – 15 minutes. Verify tokens and create authenticated sessions.

7. Phase 5 - Google OAuth

Implement OAuth using Passport and link external identities to user accounts.

8. Phase 6 - JWT and Refresh Tokens

Issue short - lived access tokens and rotating refresh tokens stored securely in HTTP - only cookies.

9. Phase 7 - Frontend Flow

Create login page, verify page, onboarding wizard, and protected dashboard.

10. Phase 8 - Security Hardening

Add rate limiting, CSRF protection, audit logs, device tracking, and suspicious login detection.

11. Phase 9 - Testing

Write unit, integration, and end - to - end tests for all authentication flows.

12. Phase 10 - Deployment

Deploy web and API, configure PostgreSQL and Redis, set environment variables, and monitor logs.

13. Recommended Timeline

Week 1: Setup and database

Week 2: Email and magic links

Week 3: JWT and sessions

Week 4: OAuth and frontend

Week 5: Security and testing

Week 6: Deployment and polishing

14. Skills You Will Learn

Authentication architecture, OAuth 2.0, JWT, Redis, queues, email systems, security, testing, and deployment.